HSDSD Practice Questions

1 Practice Questions HSDSD Midterm

1.1 Chapter 1 Interconnect Design Basics

1.1.1 Conceptual

Q1.1 *(chap1.pdf, p.3)*
Why must a trace be considered as a transmission line rather than a simple wire when its length is comparable to the wavelength of the signal?

Q1.2 *(chap1.pdf, p.4)*
What is the threshold region of a logic signal, and why is minimizing time spent in this region important for digital system design?

Q1.3 *(chap1.pdf, p.5)*
Explain the trade-off between rise/fall time and signal integrity. Why does a faster edge rate lead to more severe transmission line effects?

1.1.2 Calculation

Q1.4 *(chap1.pdf, p.6)*
A digital signal has a rise time of $\tau_r = 500$ ps. What is the highest frequency of interest?

Q1.5 *(chap1.pdf, p.7)*
A PCB trace has a propagation delay of $T_D = 200$ ps. A driver has $t_r = 1$ ns. Should this trace be treated as a transmission line?

1.1.3 True/False

Q1.6 *(chap1.pdf, p.8)*
True or False: According to Moores Law, bus speed always scales proportionally with core CPU frequency.

Q1.7 *(chap1.pdf, p.7)*
True or False: A trace of length $\lambda/20$ at the highest frequency of interest must be treated as a transmission line.

1.2 Chapter 2 Ideal Transmission Line Fundamentals

1.2.1 Conceptual

Q2.1 *(chap2.pdf, p.3)*
What are the two main planar transmission line structures used in PCB design? How do they differ in terms of layer position and effective dielectric constant?

Q2.2 *(chap2.pdf, p.6)*
List the four elements of the RLCG per-unit-length model of a transmission line and explain what physical phenomenon each one represents.

Q2.3 *(chap2.pdf, pp.3136)*
Compare series source termination and parallel load termination: describe the mechanism of each, and give one advantage and one disadvantage of each.

Q2.4 *(chap2.pdf, p.27)*
What happens to reflections when the rise time satisfies $\tau_r > 2\text{TD}$? How does this differ from the case $\tau_r < 2\text{TD}$?

1.2.2 Calculation

Q2.5 *(chap2.pdf, p.9)*
A microstrip line has $W = 4$ mil, $H = 4$ mil, $T = 1$ mil, $\varepsilon_r = 4.5$. Estimate Z_0 using the microstrip approximation formula. (Hint: check validity condition $0.1 < W/H < 2$.)

Q2.6 *(chap2.pdf, p.11)*
A stripline is embedded in FR4 ($\varepsilon_r = 4.0$) and is 3 in. long. Find the propagation velocity and time delay. ($c = 3 \times 10^8$ m/s, 1 in. = 0.0254 m)

Q2.7 *(chap2.pdf, pp.1314)*
A transmission line has $Z_0 = 75\,\Omega$, length = 4 in., $\varepsilon_r = 4.0$. A signal with rise time $T_r = 2$ ns drives the line. How many SPICE segments are needed?

Q2.8 *(chap2.pdf, p.17)*
A source of $V_s = 3.3$ V with $Z_s = 22\,\Omega$ drives a $50\,\Omega$ transmission line. What is the initial incident voltage V_i launched onto the line?

Q2.9 *(chap2.pdf, p.18)*
Calculate the reflection coefficient for: (a) a matched load, (b) an open load, (c) a load $Z_L = 100\,\Omega$ on a $50\,\Omega$ line.

Q2.10 *(chap2.pdf, pp.1516 Example 2.1)*
Given a stripline with $Z_0 = 50\,\Omega$, length $x = 5$ in., $T_r = 2.5$ ns, $\varepsilon_r = 4.5$: - (a) Calculate TD. - (b) Find L_{total} and C_{total}. - (c) How many segments are needed? - (d) Find C_{seg} and L_{seg}.

1.2.3 Multiple Choice

Q2.11 *(chap2.pdf, p.18)*
An open-circuit load on a transmission line produces a reflection coefficient of:
(A) 1 (B) 0 (C) +1 (D) +0.5

Q2.12 *(chap2.pdf, p.12)*
On the same FR4 board, which transmission line structure has a FASTER propagation speed?
(A) Stripline (B) Microstrip (C) Both the same (D) Depends only on width

---

1.3 Chapter 3 Crosstalk

1.3.1 Conceptual

Q3.1 *(chap3.pdf, p.3)*
What are the two main ways crosstalk degrades a digital bus? Name the two physical coupling mechanisms.

Q3.2 *(chap3.pdf, pp.1112)*
Compare near-end crosstalk (NEXT) and far-end crosstalk (FEXT): where is each measured, and what determines the duration of each?

Q3.3 *(chap3.pdf, pp.2432)*
Explain why odd-mode propagation results in underdriven conditions, while even-mode propagation results in overdriven conditions.

Q3.4 *(chap3.pdf, p.32)*
Why does crosstalk NOT induce velocity variations in a stripline, while it DOES in a microstrip?

Q3.5 *(chap3.pdf, pp.4142)*
Give three design rules for minimizing crosstalk between parallel traces.

1.3.2 Calculation

Q3.6 *(chap3.pdf, p.13)*
Two matched-terminated ($Z_0 = 50\,\Omega$) coupled lines have: - $L_m = 0.5$ nH/in., $C_m = 0.08$ pF/in. - $L_{11} = 8$ nH/in., $C_{11} = 3.2$ pF/in. - $V_{\text{in}} = 1$ V, $T_r = 150$ ps, line length $X = 3$ in.

First verify whether $T_r < 2\text{TD}$. Then calculate peak NEXT and FEXT amplitudes.

Q3.7 *(chap3.pdf, pp.27, 29)*
For a coupled pair of lines with $L_{11} = 7.13$ nH/in., $L_{12} = 0.54$ nH/in., $C_{11} = 2.85$ pF/in., $C_m = 0.079$ pF/in.: - (a) Calculate Z_{odd} and Z_{even}. - (b) Is the odd mode underdriven or overdriven for a $Z_s = 50\,\Omega$ source?

Q3.8 *(chap3.pdf, p.21)*
Two coupled lines have $L_{11} = L_{22} = 6$ nH/in., $L_{12} = 0.6$ nH/in. Calculate the inductive coupling factor K.

1.3.3 True/False

Q3.9 *(chap3.pdf, p.13)*
True or False: In the long-line case ($T_r < 2\text{TD}$), near-end crosstalk amplitude is independent of line length.

Q3.10 *(chap3.pdf, p.32)*
True or False: Even-mode propagation always results in a higher characteristic impedance than the single-line impedance Z_0.

Q3.11 *(chap3.pdf, p.14)*
True or False: In the short-line case ($T_r > 2\text{TD}$), far-end crosstalk equations need to be adjusted by a factor of $2\text{TD}/T_r$.

1.4 Chapter 4 Coupled Microstrip Line

1.4.1 Conceptual

Q4.1 *(chap4.pdf, p.10)*
Explain why far-end crosstalk (FEXT) exists in a microstrip but not in a stripline. Use the concept of even-mode and odd-mode time delays.

Q4.2 *(chap4.pdf, pp.2224)*
What is the purpose of the front-end decoupling capacitor C_f in coupled microstrip design? What property does it modify?

Q4.3 *(chap4.pdf, pp.3436)*
Why does adding a front-end decoupling capacitor C_f degrade eye diagram performance at high bit rates?

Q4.4 *(chap4.pdf, pp.3942)*
How do distributed decoupling capacitors improve on the lumped front-end C_f design? What is the key advantage in terms of S-parameters?

1.4.2 True/False with Explanation

Q4.5 *(chap4.pdf, p.10)*
True or False: FEXT is always zero in a stripline because the dielectric fully surrounds both conductors, making even- and odd-mode velocities equal.

Q4.6 *(chap4.pdf, p.43)*
True or False: Adding a front-end decoupling capacitor reduces both FEXT and NEXT simultaneously.

1.5 Chapter 5 Non-ideal Interconnect Issues

1.5.1 Conceptual

Q5.1 *(chap5.pdf, p.6)*
Define skin depth δ. What physical phenomenon causes the skin effect, and what happens to conductor resistance as frequency increases?

Q5.2 *(chap5.pdf, p.7)*
Explain why the internal inductance of a conductor decreases at high frequencies. What is the asymptotic high-frequency value?

Q5.3 *(chap5.pdf, pp.1719)*
Describe the two visual effects of AC conductor losses on a time-domain digital waveform.

Q5.4 *(chap5.pdf, pp.3133)*
Why do serpentine traces cause ledges on the rising edge of a signal? What design rules mitigate this?

Q5.5 *(chap5.pdf, pp.3436)*
What is intersymbol interference (ISI) and under what condition is it most severe?

1.5.2 Calculation

Q5.6 *(chap5.pdf, p.6)*
Copper has $\rho = 1.72 \times 10^{-8}$ ům, $\mu = \mu_0 = 4\pi \times 10^{-7}$ H/m. Calculate the skin depth at 1 GHz.

Q5.7 *(chap5.pdf, p.15)*
What is the surface resistance R_S of copper? (Use $\rho = 1.72 \times 10^{-8}$ ům.)

Q5.8 *(chap5.pdf, p.12)*
A microstrip has $W = 5$ mil $= 127\,\mu$m, $H = 3.2$ mil $= 81.3\,\mu$m. Estimate R_{ac} at 1 GHz using the approximate formula. ($\rho_{\text{Cu}} = 1.72 \times 10^{-8}$ ům, $\mu = \mu_0$)

Q5.9 *(chap5.pdf, p.21)*
A microstrip has $C_{11} = 3.2$ pF/in. and the FR4 substrate has $\tan\delta = 0.02$. Calculate the shunt conductance G per unit length at 1 GHz.

Q5.10 *(chap5.pdf, p.39)*
A microstrip trace with $C_{11} = 3.5$ pF/in. and $w = 5$ mil makes a 90° bend. Estimate the excess capacitance.

1.5.3 True/False

Q5.11 *(chap5.pdf, p.6)*
True or False: Skin depth increases with frequency.

Q5.12 *(chap5.pdf, p.33)*
True or False: Clock traces are generally acceptable to serpentine because the effect is the same on all edges.

Q5.13 *(chap5.pdf, p.40)*
True or False: Chamfering a 90° bend at 45° eliminates excess capacitance.

---

2 Answers

2.1 Chapter 1

A1.4: $f_{\max} = 0.32/500$ ps $= 0.64$ GHz

A1.5: $t_r/10 = 100$ ps. Since $T_D = 200$ ps > 100 ps, YES treat as a transmission line.

A1.6: False. Bus speed tends to lag behind core frequency (bus is the bottleneck). *(chap1.pdf, p.9)*

A1.7: False. Only conductors $\geq \lambda/10$ need to be treated as TX lines; $\lambda/20$ is safely below the criterion. *(chap1.pdf, p.7)*

2.2 Chapter 2

A2.5: $W/H = 4/4 = 1.0$ (valid). $Z_0 \approx \frac{87}{\sqrt{4.5+1.41}} \ln \frac{5.98\times4}{0.8\times4+1} = \frac{87}{\sqrt{5.91}} \ln \frac{23.92}{4.2} \approx 35.8 \times \ln(5.695) \approx 35.8 \times 1.739 \approx 62.2\,\Omega$

A2.6: $v = 3 \times 10^8/\sqrt{4} = 1.5 \times 10^8$ m/s; TD $= 3 \times 0.0254/1.5 \times 10^8 = 508$ ps

A2.7: TD: $v = 3 \times 10^8/\sqrt{4} = 1.5 \times 10^8$; TD $= 4 \times 0.0254/1.5 \times 10^8 = 677$ ps. Segments $\geq 10 \times 677$ ps/2000 ps $= 3.4 \to 4$ segments.

A2.8: $V_i = 3.3 \times 50/(50 + 22) = 3.3 \times 0.694 = 2.29$ V

A2.9: (a) $\rho = 0$; (b) $\rho = +1$; (c) $\rho = (100 - 50)/(100 + 50) = +0.333$

A2.10: - (a) TD $= 5 \times 0.0254 \times \sqrt{4.5}/(3 \times 10^8) = 898$ ps - (b) $L_{\text{total}} = 898$ ps $\times\, 50\,\Omega = 44.9$ nH; $C_{\text{total}} = 898$ ps$/50\,\Omega = 17.9$ pF - (c) Segments $\geq 10 \times 898/2500 = 3.6 \to 4$ - (d) $C_{\text{seg}} = 17.9/4 = 4.48$ pF; $L_{\text{seg}} = 44.9/4 = 11.23$ nH

A2.11: (C) $+1$ (open circuit: $Z_t \to \infty$, $\rho \to +1$) *(chap2.pdf, p.18)*

A2.12: (B) Microstrip. Partial air smaller effective ε_r faster propagation. *(chap2.pdf, p.12)*

2.3 Chapter 3

A3.6: - TD $= X\sqrt{L_{11}C_{11}} = 3 \times \sqrt{8 \times 10^{-9} \times 3.2 \times 10^{-12}} = 3 \times \sqrt{25.6 \times 10^{-21}} = 3 \times 160$ ps $= 480$ ps - 2TD $= 960$ ps. $T_r = 150$ ps < 960 ps long-line case - $A_{\text{NEXT}} = (1.0/4)(0.5/8+0.08/3.2) = 0.25(0.0625+0.025) = 0.25 \times 0.0875 = 21.9$ mV - $B_{\text{FEXT}} = -(1.0 \times 3 \times 160 \text{ ps})/(2 \times 150 \text{ ps}) \times (0.0625 - 0.025) = -(480/300) \times 0.0375 = -1.6 \times 0.0375 = -60$ mV

A3.7: - $Z_{\text{odd}} = \sqrt{(7.13 - 0.54)/(2.85 + 0.079)} = \sqrt{6.59/2.929} = \sqrt{2.251} \approx 50.0/\sqrt{2.251} \cdots$

More directly: $Z_{\text{odd}} = \sqrt{6.59 \text{ nH/in.}/2.929 \text{ pF/in.}} = \sqrt{2251\,\Omega^2} \approx 47.4\,\Omega$

- $Z_{\text{even}} = \sqrt{(7.13 + 0.54)/(2.85 - 0.079)} = \sqrt{7.67/2.771} = \sqrt{2.768 \times 10^3} \approx 52.6\,\Omega$
- For $Z_s = 50\,\Omega$: odd mode has $Z_{\text{odd}} = 47.4\,\Omega < Z_s = 50\,\Omega$ **underdriven** *(chap3.pdf, p.27)*

A3.8: $K = L_{12}/\sqrt{L_{11}L_{22}} = 0.6/\sqrt{6 \times 6} = 0.6/6 = 0.10$ *(chap3.pdf, p.21)*

A3.9: True. *(chap3.pdf, p.14)*

A3.10: True. $Z_{\text{even}} > Z_0$ always. *(chap3.pdf, p.29)*

A3.11: False. It is NEXT (not FEXT) that is adjusted by $2\text{TD}/T_r$ in the short-line case. FEXT equations do not need adjustment. *(chap3.pdf, p.14)*

2.4 Chapter 4

A4.5: True. In a stripline both conductors are fully surrounded by the same dielectric; the effective ε_r is the same for both modes equal velocities $T_d = 0$ FEXT cancels. *(chap4.pdf, p.10)*

A4.6: False. Adding front-end C_f **reduces FEXT** but **increases NEXT**. *(chap4.pdf, pp.2627)*

2.5 Chapter 5

A5.6: $\delta = \sqrt{1.72 \times 10^{-8}/(\pi \times 10^9 \times 4\pi \times 10^{-7})} = \sqrt{1.72 \times 10^{-8}/(1.974 \times 10^{-3})} = \sqrt{8.71 \times 10^{-6}} \approx 2.95\,\mu\text{m}$

A5.7: $R_S = \sqrt{\rho\pi\mu} = \sqrt{1.72 \times 10^{-8} \times \pi \times 4\pi \times 10^{-7}} = \sqrt{1.72 \times 10^{-8} \times 3.948 \times 10^{-6}} = \sqrt{6.79 \times 10^{-14}} \approx 8.24 \times 10^{-7}\,\Omega \cdot \sqrt{\text{s}}$

A5.8:

$$R_{\text{ac}} = \sqrt{1.72 \times 10^{-8} \times \pi \times 4\pi \times 10^{-7} \times 10^9}\left(\frac{1}{127 \times 10^{-6}} + \frac{1}{6 \times 81.3 \times 10^{-6}}\right)$$

$= \sqrt{0.068} \times (7874 + 2049) = 0.261 \times 9923 \approx 2590\,\Omega/\text{m}$

A5.9: Convert C_{11}: 3.2 pF/in. $\times$ 39.37 in./m $= 126$ pF/m.
$G = 0.02 \times 2\pi \times 10^9 \times 126 \times 10^{-12} = 0.02 \times 0.792 = 0.01584$ S/m ≈ 15.8 mS/m

A5.10: $C_{90^\circ} \approx C_{11} \times w = 3.5$ pF/in. $\times$ 5 mil $= 3.5 \times (5/1000)$ pF $= 0.0175$ pF ≈ 17.5 fF
(Note: units must be consistent w in inches if C_{11} is in pF/in.)

A5.11: False. Skin depth **decreases** with increasing frequency. *(chap5.pdf, p.6)*

A5.12: False. Clock traces must **never** be serpentined because early signal arrival corrupts clock timing. *(chap5.pdf, p.33)*

A5.13: False. Chamfering **reduces** excess capacitance but does not eliminate it completely. Rounding is more effective, and 45° bends avoid the issue altogether. *(chap5.pdf, p.40)*